

University Degree in Telematics Engineering
2024-2025

Bachelor Thesis

“Machine Learning-Based Predictive Modeling of Energy Prices”

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SUMMARY

This project focuses on the development of a machine learning-based predictive model for electricity prices in Spain. Using historical data from the OMIE and technical analysis (TA) indicators, the model aims to accurately forecast hourly energy prices. The study focuses on a single hourly slot, evaluating the performance of various Machine Learning models, such as linear regression, Lasso, and Random Forest. There was a strong focus on the feature engineering, employing technical analysis indicators such as moving averages, exponential moving averages, and momentum metrics. The results display the impact of tailoring the features to improve model accuracy and offer insights into the potential of data-driven approaches for energy price forecasting.

Keywords:

Energy

Machine Learning

Sliding Window

Technical Analysis (TA)

DEDICATION

Dedicated to my late grandfather, who was not able to see me become an engineer like himself.

To my amazing family.

To my parents, that have always helped me push through hard moments.

To my grandparents who will proudly see me become an engineer.

To my amazing girlfriend that has supported me though all the ups and downs of life.

To my all of friends and colleagues that have accompanied me these years.

To anyone and everyone that has supported me during my years in university.

Here is to the next steps in life

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1. INTRODUCTION TO THE PROBLEM

- **1. Explanation of how the price of energy works in Spain**
 - a. Energy
 - b. Prices
- **2. Machine Learning - Lets run through some model overviews, which are available to choose, but without actually referencing anything**

Energy has increasingly become more important. Availability is key for a modern high functioning society. In Europe we are lucky to have a really advanced and reliable network. It is a key resource needed to succeed in the ecological transition that we are currently undergoing world wide. Electric mobility is a key element of this process. Electric cars, buses and trucks, need easily available and convenient sources of energy to recharge, to be able to play their role in society. An important factor in this ever growing necessity is the price of the available energy. But in recent times, that easy and affordable access has changed, and has become more unpredictable. Ever since the commencement of the war between Russia and Ukraine marked an inflection point in the mostly cyclical nature of electricity prices. Prices have gone up considerably since a couple years time, and it has been affecting everyone, from big consumers such as companies, to everyday people, having to pay a more expensive electricity bill at the end of the month. This has become a topic of great concern, with more and more people thinking about it often.

This led me to think about pricing, how it was structured, and what to expect as an end consumer. Could we plan our use of energy according to the price? Could it be predicted accurately? These were some of the questions I had that sparked my interest in looking into energy predictions.

In the initial discussions of this project, we talked about various ways to attack the problem at hand. We thought about a variety of systems but settled on a plan to explore the predictions with the more established methods of Machine Learning. This was done specially since it was a natural continuation of what was learned in the third year subject of Telematics Engineering, Modern Theory of Detection and Estimation.

The first idea we developed was to separate out the project in 24 blocks, one for each hour block to predict. The reason behind splitting the data in 24 blocks was simple, we assumed that data between the same time slot across neighbor days, would be similar, or would follow a certain trend.

Focusing on the innovation side of things, we decided to simplify the problem. Pivoting from creating an algorithm to predict hourly prices, to splitting that into the 24 blocks, and predicting a single one. We decided on the 14th hour of each day, since irrespective of the day of the week, or holiday, it would always be a valley period.

This kind of granular breaking up of data is similar to what a Random Forest does in order to absorb more details and create higher resolution predictions. We made this assumption based on the idea that it would simplify the project substantially.

Talk about Emilio's background in investment banking at BBVA and his algorithms.

Talk about the idea of using technical analysis for feature engineering

2. BACKGROUND AND RELATED WORK

2.1. General background

How is the price of electricity set in Spain and Portugal?

LEER ARTICULO [1]

What happens when new producers enter the market and sell to the energy pool?

Comment about more typical case studies focusing on energy consumption, since that is something more consistent and predictable

2.2. The Sliding Window plan: Setting up the Feature Matrix

How do I prepare my data to utilize across different models: The Feature Matrix (weight matrix)

The sliding window theory (mini-models)

2.3. Which Machine Learning models will be used

What algorithms will I use?

- Linear Regression: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html

- Lasso Regression: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.Lasso.html

- Random Forest: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html>

2.4. Fine Tuning with Technical Analysis and Feature engineering

What is *Fine-tuning*? - <https://www.ibm.com/es-es/think/topics/fine-tuning>

What Type of Fine-tuning will I use?

- Feature Engineering & Feature Selection

- Technical analysis based feature engineering

What is Technical Analysis? - <https://www.britannica.com/money/what-is-technical-analysis>

TA metrics used? SMA, EMA, ROC & RSI

Talk about the intervals used:

$SMA_3, SMA_5, SMA_7, SMA_{14}, SMA_{30}, SMA_{60}, SMA_{90}, SMA_{180}, SMA_{360}$

$EMA_3, EMA_5, EMA_7, EMA_{14}, EMA_{30}$

$ROC_3, ROC_5, ROC_7, ROC_{12}, ROC_{14}, ROC_{30}$

RSI_5, RSI_7, RSI_{14}

2.5. Evaluation Metrics

How will I measure my accuracy?

errors:

R^2

RMSE & MSE

2.6. Related Work

Other ways of predicting

Temporal Series

Energy consumption predictions

3. PROPOSAL: THEORETICAL SYSTEM DESCRIPTION

This chapter will outline the design of the system developed for the project, describing the complete pipeline constructed for the system. This will be explained starting from a blank state, detailing the complete set-up process required to run the project. Following this initial configuration, the data preparation stage will be outlined. This will be continued by the feature matrix creation process, concluding with the training of the machine learning models and the chosen evaluation methods.

The following Block diagram will model in the most basic forms the system design:

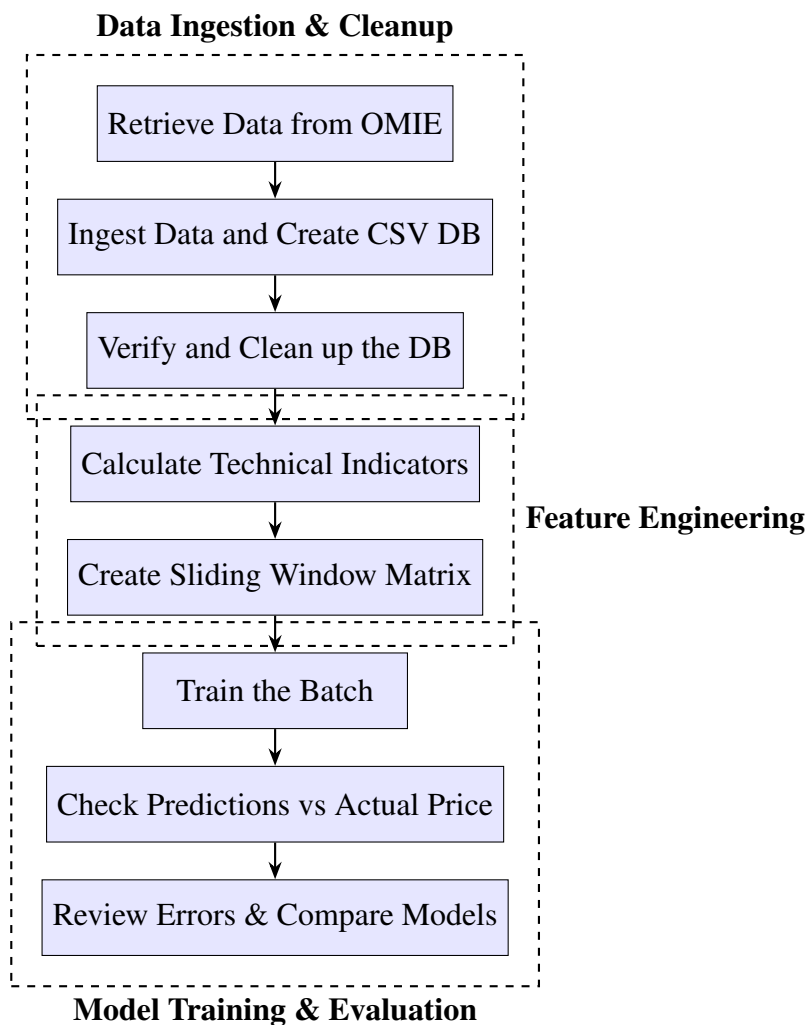


Fig. 3.1. Theoretical Block Diagram of the Project Pipeline

3.1. Prior Set Up and Configuration

Before getting into the system explanation, there are some important pre-requisites that must be met in order to run the project. The project was developed and tested with Python 3.13 [2], so for any recreation of the project, it is highly recommendable to use the same version. For additional assurance of reproducibility, and as a universal best practice for Python development, it is strongly advisable to set up a new virtual environment [3]. Alternatively, another system that would achieve the same goal of reproducibility, and dependency conflict avoidance would be the use of Miniconda [4]. In this project we chose to use the built in Python module *venv*, with which a virtual environment can be easily created and configured with the project dependencies using the following commands.

Create the virtual environment:

```
python -m venv .venv
```

Activating the new virtual environment:

in macOS

```
source .venv/bin/activate
```

in Windows

```
.venv\Scripts\Activate.ps1
```

To reproduce the exact environment used in the project, a *requirements.txt* file must be created with the following contents:

```
ipykernel==6.29.5
matplotlib==3.10.1
numpy==2.2.3
pandas==2.2.3
requests==2.32.3
scikit-learn==1.6.1
ta==0.11.0
```

To install the dependencies run:

```
pip install -r requirements.txt
```

Following the completion of the prior set up required for the project, the following sections will detail in depth the different steps involved in the system pipeline.

3.2. Data Preparation: Ingestion and Cleanup

3.3. The Feature Matrix Creation

3.4. Model Training

Data Preparation

Matrix Creation

Mini-models - sliding window approach

The thought process behind our sliding window approach has to do with the data and the

No estacionario.

No son datos IID - Independientes e idénticamente distribuidos.

Hay correlación en mis datos entre cada día? Si debería ser relativamente alta, estacional, pero progresiva y lenta.

Technical indicators

Que es cada cosa y porque usarlos en vez de series temporales, porque la info temporal esta implicita en los indicadores. Indicadores que capturan patrones en la serie Ventaja: modelo ML mas sencillo pq las features son muy informativas vs

ML blocks Linear Regression Linear combination of every feature Lasso Regression Can turn unimportant features into 0s Random Forest Feature engineering

Pandas, NumPy, scikit-learn

Retrieve Data from OMIE

Kick-starting the pipeline design, the data is retrieved from the OMIE website, and is ingested into a CSV file that will act as our rudimentary database. The process is done with the following code:

After a successful retrieval, the database is checked for missing or repeated data points. This will be checked in a time-slot basis.

Since the idea is to cater to any potential system, it was created without any hour slot preference, but in the system I will be proposing, we will limit the dataset to just the 14h time-slot.

The database will be transformed into a reduced set, containing only the relevant data points, measured always at 14h

Subsequently comes the creation of the feature matrix. This will initially consist of exclusively the sliding window width desired for that training batch.

Once the matrix is created (and verified), the training process begins.

Due to our particular system, the training will consist of complete rows of that feature matrix.

Cross validation?? Not really done, in another way w the sliding window cambio parametros pero no la arquitectura del modelo

Review the mini model theory, why is it reasonable to not use training set, validation and test sets?

Review error rates and compare with other iterations/ batches and other models.

Detailed Steps:

1. Download of OMIE data (only section independent of the whole system) [5]

Downloaded with a Python script for the dates that were not available with an easy .zip file (for whole previous years):

Insert the python script?? better suited for an annex??

```
1 import requests
2 from datetime import datetime, timedelta
3
4 # Usage instructions:
5     # Set file name to save the filenames to download
6     # Set start and end dates
7
8 file_path = 'to_download.txt'
9 start_date = datetime(2025, 2, 14)
10 end_date = datetime(2025, 3, 18)
11
12 # Compose filenames to download
13 def compose_filenames():
14     start_date = start_date
15     end_date = end_date
16
17     # Generate the entries for the number of days comprehended
18     to_generate = (end_date - start_date).days + 1
19     entries = []
20     for i in range(0, to_generate): # x days from start to end
21         date_entry = start_date + timedelta(days=i)
22         entries.append(f'marginalpdbc_{date_entry.strftime("%Y%m%d")}
23             }.1')
24
25     # Save to a .txt file
26     with open(file_path, 'w') as file:
27         for entry in entries:
28             file.write(entry + '\n')
29
30 # Example URI to build
31 # https://www.omie.es/es/file-download?parents%5B0%5D=marginalpdbc&
```

```

filename=marginalpdbc_20230102.1
31 def read_filenames_and_compose_urls(file_path):
32     base_url = "https://www.omie.es/es/file-download?parents%5B0%5D=
        marginalpdbc&filename="
33
34     # Read the filenames from the .txt file
35     with open(file_path, 'r') as file:
36         filenames = [line.strip() for line in file if line.strip()]
37
38     # Compose the URLs
39     urls = [base_url + filename for filename in filenames]
40
41     return urls
42
43 def download_files(urls):
44     for url in urls:
45         # Extract the filename from the URL
46         filename = url.split('=')[-1]
47
48         # Send a GET request to download the file
49         # response = requests.get(url)
50         response = requests.get(url)
51
52         # Check if the request was successful and write to file
53         if response.status_code == 200:
54             # Write the content to a file with the extracted
                filename
55             with open(filename, 'wb') as file:
56                 file.write(response.content)
57             print(f"Downloaded {filename}")
58         else:
59             print(f"Failed to download {filename}")
60
61 # Execution
62 compose_filenames()
63 urls = read_filenames_and_compose_urls(file_path)
64 download_files(urls)

```

2. Data shape? NEXT CHAPTER 4

3. Data format: NEXT CHAPTER 4

4. Data organization

We will transform the data into a CSV format easier to use with the Pandas library to later create a DataFrame. We will maintain the same information, but in a encoding more suitable for Pandas. For that, the date and time must be transformed to a format such as Datetime which will transform the information from 2018;01;01;14; ... to 2018-01-01 14:00:00

5. Data visualization with Seaborn? With Microsoft Corporation's "Data Wrangler"

VSCode extension

6. Data retention of only the relevant info time and date, and MarginalES - drop MarginalPT

7. Data cleanup: Check for missing or duplicate entries. since Daytime Savings introduces duplicate hours some days of the year: INSERT some dates in the dataset that have the duplicated entries

8. Explain all blocks of the system? Data download: Data ingest

Contar los bloques empleados en cada parte – como junto las piezas para construir el sistema – porque uso las piezas que uso

Hablar de implementaciones como que librerías, que scripts etc – no es muy necesario, mas importante la idea

4. EXPERIMENTATION

This chapter is dedicated to reviewing all of the relevant information pertaining the experimentation phase of the project. We will review the specifics of the data that was dealt with, and explain the decisions taken for the preparation of such data, and the training of the different machine learning models employed. We will review all of the results of the different models, and discuss in depth the different set-up variations for the models.

4.1. Data Description

The data that was used for this project was exclusively publicly available information, published by the OMIE, which stands for *Operador del Mercado Ibérico de Energía*, meaning the Iberian operator of the energy market. The data that was retrieved for the project takes the following shape:

```
MARGINALPDBC;  
2018;01;01;1;28.1;6.74;  
2018;01;01;2;33;4.74;  
2018;01;01;3;32.9;3.66;  
2018;01;01;4;28.1;2.3;  
2018;01;01;5;27.6;2.3;  
2018;01;01;6;24.6;2.06;  
2018;01;01;7;20.1;2.06;  
2018;01;01;8;19.9;2.06;  
2018;01;01;9;19.84;2.3;  
2018;01;01;10;19.9;2.3;  
2018;01;01;11;19.9;2.3;  
2018;01;01;12;19.9;2.3;  
2018;01;01;13;23.6;2.3;  
2018;01;01;14;25.1;2.3;  
2018;01;01;15;23.6;5;  
2018;01;01;16;24.6;5;  
2018;01;01;17;25.1;5;  
2018;01;01;18;27.6;8.85;  
2018;01;01;19;27.6;15.93;  
2018;01;01;20;28.1;22.02;  
2018;01;01;21;32.9;20;  
2018;01;01;22;28.1;21.95;  
2018;01;01;23;28.1;23.52;  
2018;01;01;24;27.6;16.35;
```

*

We can understand and interpret this format following the OMIE's guide: *Modelo de Ficheros para la distribución pública de Información del mercado de electricidad 1.35* [6]. In page number 67, chapter 6.18 we may find the following information regarding the encoding format for the information:

6.18 Precios marginales del mercado diario (MARGINALPDBC)

Fichero con los precios marginales del Mercado Diario para cada una de las horas.

Nombre del fichero: `marginalpdbc_aaaammdd.v` donde `aaaammdd` corresponde a la fecha de sesión y `v` es la versión del fichero.

Descripción de los campos:

CAMPO DESCRIPCIÓN VALORES VÁLIDOS

Año Año I4 - 20XX

Mes Mes I2 - 1 a 12

Día Día I2 - 1 a 31

Hora Hora I2 - 1 a 25

MarginalPT Precio marginal zona Portuguesa F8.2 - -99999.99 a 99999.99

MarginalES Precio marginal zona Española F8.2 - -99999.99 a 99999.99

From this data description we can obtain the following column names: *Año*, *Mes*, *Día*, *Hora*, *MarginalPT* and *MarginalES*. As a proof of concept, in this investigation we will be focusing on a single time slot, simplifying the data, from multiple data points per day, to a single one. We are exclusively interested in the value of the last column, *MarginalES*, and specifically the row pertaining the 14th hour of each day, which we will refer to as 14H.

For the data visualization portion, I employed various methods, both plotting values with matplotlib, seaborn and using the Microsoft Data Wrangler Visual Studio Code extension.

Explain some more about the data distribution.

Explain the ups and downs of the data (war and relation to natural gas prices).

Explain the zero price.

4.2. Set Up Explanation

How do I use the previously explained system - what parameters did I modify to obtain the final results. Feature engineering, Alpha variation testing in Lasso, Tree depth and number of leafs. Que modelos y que configs

- Linear Regressor

- Lasso Regressor
- Random Forest

4.3. Results of the Testing

Results as graph/ tables - This would be to compare a model across various iterations. Model best-run comparisons, etc.

Error medio en todas las predicciones Y percentiles (expectation shortfall tambien?)

4.4. Low Level Discussion

This would be the final explanation that I would give to a colleague of mine, with full details on how I have done everything

5. REGULATORY FRAMEWORK

Not essential for this investigative project because we do not go to market.

5.1. Data Availability

Habria que hablar de las normas, pq si alguien lo usa es para ir a mercado

Current Spanish and European regulations allow for the use of publicly available institutional data for use in educational investigative work.

I need to talk about laws, what data is permissible to use and how. This would be essential for a project that does indeed go to market.

For us, its not that relevant.

5.2. Software & Licenses

For the development of this project I have used a variety of open and closed source utilities and code.

For the sake of organization, I will categorize this section into two distinct parts, required software to reproduce the project, and optional software for ease of production, plannification, organization or otherwise related.

Required:

Python - This was the programming language in wich the entirety of the project was developed [7]

Scikit-learn - library for ml [8]

Pandas - data management [9]

ta library - Technical Analysis library[10]

Light use of Seaborn - data visualization [11]

NumPy - number manipulation [12]

macOS / Windows - closed source platforms across where the project was developed [13] & [14]

Optional:

Git - a distributed version control system [15]

GitHub - Repository hosting platform [16]

Visual Studio Code - open source text editor for my general coding needs [17]

Visual Studio Code extensions such as Microsoft Data Wrangler, Python packages or such

Python Virtual Environment (venv) - This was created as it is best practice to create new Python virtual environments for each new project, as specially in a production environment these usually require certain specific versions that must be met in order for all components to work as intended. An alternative method of dealing with this would be using Conda, and creating with it a new Conda environment, [3]

Overleaf - closed source platform used to compile LaTeX code [18]

LaTeX - open source language used to create the final project document [19]

OpenAI ChatGPT, Google Gemini & Anthropic Claude - closed source large language models used to aid in troubleshooting general coding issues

6. SOCIO-ECONOMIC ENVIRONMENT

6.1. Socio-economic Impact

Why is this relevant?

- Energy Prediction
- Shutdowns
- Price sensitivity

6.2. Project plan

TODO: Gantt diagram representing time spent in each phase - Design and innovation time frames

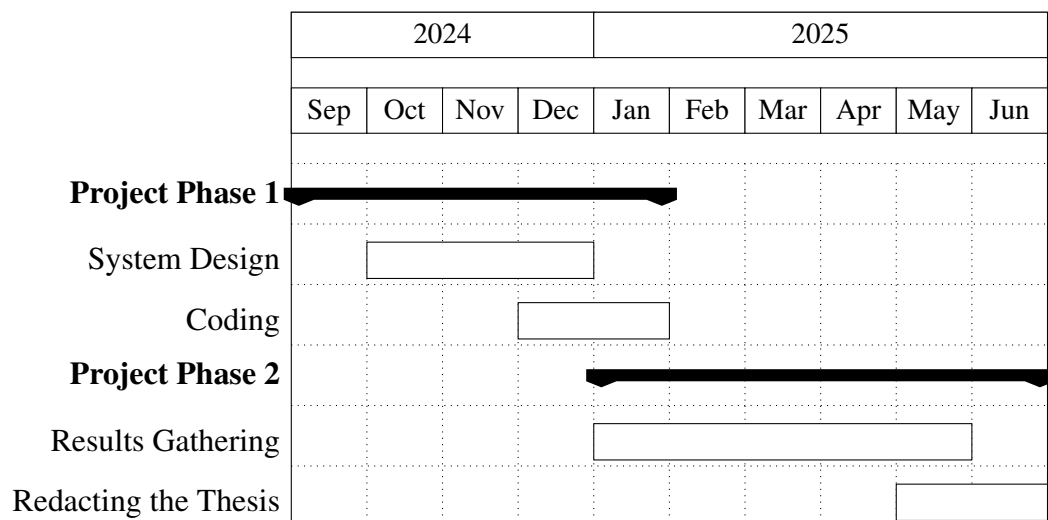


Fig. 6.1. Gantt Diagram of the Project Lifecycle

6.3. Budget

This section will consist of a complete price breakdown of the investigation, consisting of material costs such as the equipment utilized, and the human capital that composed the investigation team. It is notable to highlight that as previously mentioned, no paid license of any sort was obtained for the realization of this project. All of the programs, tools or code, were either *Open Source* or free to use software under exclusive *non-commercial use* licenses.

The most basic requirement for the physical needs of this project is a computer with

an desktop operating system. Since the project was written in Python [7], it would be reasonable to consider it platform agnostic. Any computer capable of running any recent operating system, configured with a recent version of Python would be able to run this project. This includes any of the three most popular desktop operating systems, Windows, macOS and any Linux distribution.

The project was mainly tested using Python 3.13 [2] which does require a recent operating system such as macOS 13 or Windows 10. But as for the project itself, there is no platform, processor architecture, or processing power requirements. In my case, I mainly developed the code in my personal laptop, a 2021 MacBook Pro configured with the ARM M1 Pro chip and 16GB of RAM, running the latest available macOS version at this time, macOS 15 [13]. This laptop was obtained from a second-hand store circa June 2024, mainly for personal use, but also helping keep the theoretical project costs down.

The following table outlines the specific equipment costs incurred for the development of this investigative project:

Equipment	Price (€)	Amortization per year (€)	Useful Life (years)	Usage Time (months)	Cost (€)
MacBook Pro M1 Pro	1250	312.5	4	9.5	247.4

TABLE 6.1. EQUIPMENT AMORTIZATION

Regarding the human capital that was required for the successful development of this project, it will consist exclusively on two people. The director of my bachelor's thesis, professor Emilio Parrado Hernández, PhD, as the expert Machine Learning consultant, and myself as a junior software engineer.

The following table displays the estimated hourly rates of each person, the hours dedicated to the project, and the total cost:

Name	Role	Price (€/hour)	Hours	Cost (€)
Emilio Parrado Hernández	Expert ML Consultant (PhD)	200	-	-
Rodrigo De Lama Fernández	Junior Engineer (Pre-Graduate)	30	-	-

TABLE 6.2. HUMAN COSTS

For our final budgeting considerations, other miscellaneous expenses, such as transportation costs to the university for on site meetings with my bachelor's thesis director.

Summing up all costs, the following table estimates what this project would have cost to investigate:

Concept	Cost (€)
Direct/Material Costs	247.4
Engineering Costs	-
University Carlos III Costs - Expert ML Consultant	-
Total	-

TABLE 6.3. FINAL COST BREAKDOWN

7. CONCLUSIONS

In this investigation we have analyzed in depth an innovative methodology for predictive modeling of energy prices with Machine Learning algorithms, enhanced by the usage of technical analysis indicators in feature engineering, in order to enhance prediction precision. This final chapter will review the general conclusions attained throughout the development of this project's Machine Learning based solution for energy price predictions. It will also discuss potential development paths for future work pertaining the system designed in this investigative project

7.1. Review of the Investigation

After finishing the whole project, read it, and reintroduce the objectives to the reader. Remind them of the completion of them

7.2. Revisiting the Objectives

Predicting the prices in an accurate manner using Machine Learning algorithms and technical analysis

7.3. Future work

- Less absolute error
- Better overall precision: better percentile accuracy

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